FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. The position vector of a particle moving in space is given by $\vec{r} = (1+ 2\cos 2\omega t)\hat{i} + (3\sin^2\omega t)\hat{j} + (3t)\hat{k}$ in the ground frame. All the units are in SI. Choose the correct statement (s):

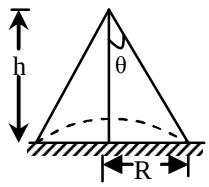
   (A) The particle executes SHM in the ground frame about the mean position $\left(1, \frac{3}{2}, 3t\right)$

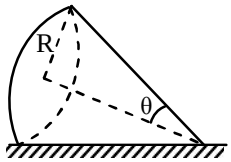
   (B) The particle executes SHM in a frame *S* moving along the positive z–axis with a velocity of 3 m/s.

   (C) The amplitude of the SHM of the particle in frame *S* is $\frac{5}{2}m$.

   (D) The line of the SHM of the particle is parallel to the vector $\left(\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}\right)$.

2. A solid cone is placed on a horizontal surface with height '*h*' and radius '*R*'. Its apex angle is 'θ'. To change the position of the cone from figure (A) to figure (B), no work is required, then





   (A) $\theta = \sin^{-1}\left(\frac{1}{3}\right)$

   (B) $\theta = \frac{\pi}{4}$

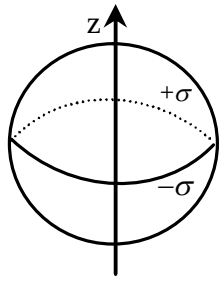
   (C) $\frac{h}{R} = 1$

   (D) $\frac{h}{R} = 2\sqrt{2}$

Space for Rough work

3. A standing wave is setup in a string fixed at both ends. Then
   (A) The sum of total energy per unit length at nodes and antinodes is constant
   (B) The total energy of a point midway between a node and an adjacent antinode is constant
   (C) If the string has a standing wave and a component of travelling wave, then the kinetic energy of the points of minimum amplitude is non-zero at some instant of time.
   (D) If the string has a standing wave and a component of travelling wave, then the potential energy of the points of maximum amplitude is non zero at some instant of time.

4. A hollow, insulating spherical shell has a surface charge distribution placed upon it, such that the upper hemisphere has a uniform surface charge density $+\sigma$, while the lower hemisphere has a uniform surface charge density $-\sigma$,as shown in the figure. Their interface lies in *x-y* plane. Which of the following statement(s) is/are correct?

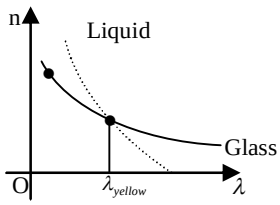


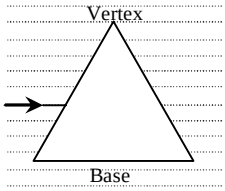
   (A) The field at all points of *x-y* plane within the sphere points in the –ve *z*-direction
   (B) All points of the *x-y* plane within the sphere are equipotential
   (C) The field at all points on *z*-axis outside the sphere point along positive *z*-direction
   (D) The field at points on *z*-axis which are on either side of origin outside the sphere is in opposite directions

5. A planoconvex lens $(\mu = 1.5)$ of focal length 20 cm has its plane side silvered. Which of the following statements is/are correct?
   (A) The radius of curvature of its curved surface is half that of a surface of equiconvex lens of focal length 20 cm made of same material
   (B) An object placed at 15 cm on the axis on the convex side gives rise to an image at a distance of 30cm from it
   (C) An object placed at a distance of 20 cm on the axis on the convex side gives rise to an image at 40cm from it
   (D) It acts as a convex mirror

Space for Rough work

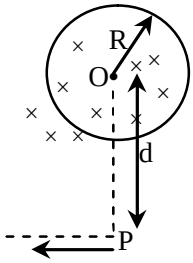
6. A glass prism is immersed in a hypothetical liquid. The curves showing the refractive index *n* as a function of wavelength λ for glass and liquid are as shown in the figure. When a ray of white light is incident on the prism parallel to the base:





(A) Yellow ray travels without deviation
(B) Blue ray is deviated towards the vertex
(C) Red ray is deviated towards the base
(D) There is no dispersion

7. In a cylindrical region of radius *R*, there exists a time varying magnetic field *B* such that $\frac{dB}{dt} = (>0)$ .A charged particle having charge *q* is placed at the point *P* at a distance $d\left(>R\right)$ from its centre *O*. Now, the particle is moved in the direction perpendicular to *OP* (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.



(A) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 2R$

(B) Work done by external agent is $\frac{q\pi R^2}{8}k$ if $d = 4R$

(C) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 4R$

(D) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 6R$

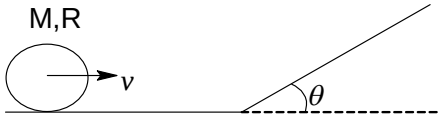
Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

An object of mass M and radius R is pure rolling on a horizontal surface. It smoothly climbs over an rough incline with sufficient friction to ensure no slipping. The following grid has 3-columns: Column- 1 represents the type of object. Column -2 represents the maximum height attained by the object and column -3 represents the time taken to reach the maximum height.



Column –1 (Object)		Column –2 (Height)		Column –3 (Time)	
(I)	Ring	(i)	$\frac{7v^2}{10g}$	(P)	$7v/5g\sin\theta$
(II)	Disc	(ii)	$\frac{3v^2}{4g}$	(Q)	$2v/g\sin\theta$
(III)	Hollow sphere	(iii)	v^2/g	(R)	$3v/2g\sin\theta$
(IV)	Solid sphere	(iv)	$\frac{5v^2}{6g}$	(S)	$5v/3g\sin\theta$

8. Pick the correct combination:-

(A) $(\mathrm{I})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{III})\ (\mathrm{ii})\ (\mathrm{Q})$

(C) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$

(D) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{P})$

Space for Rough work

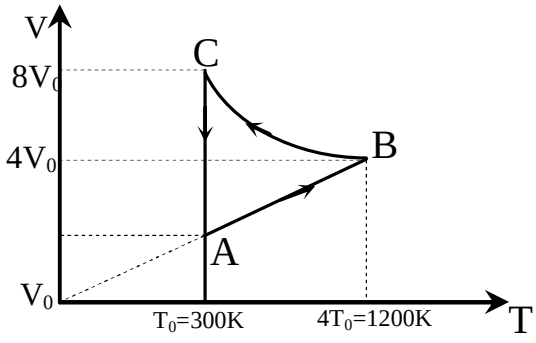
9. The combination where the total mechanical energy is maximum will be
(A) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$
(B) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$
(C) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{P})$
(D) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{S})$

10. The combination for which maximum friction acts between the body and the incline is
(A) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$
(B) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$
(C) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{S})$
(D) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{P})$

Space for Rough work

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

Two moles of an ideal mono-atomic gas is taken through a cyclic process ABCA as shown in the V-T diagram. In the process BC, TV_2 = constant. Answer the questions by referring to the data in the table. (Take $\ell n(2) = 0.7$)



Column –1 shows the processes. Column –2 shows the work done in that process and Column –3 shows the energy supplied in that process.					
	Column–1		**Column–2**		**Column–3**
(I)	$A \to B$	(i)	1800 R	(P)	-1800 R
(II)	$B \to C$	(ii)	-1260 R	(Q)	-3060 R
(III)	$C \to A$	(iii)	900 R	(R)	4500 R
(IV)	$B \to C \to A \to B$	(iv)	1440 R	(S)	1440 R

11. Pick a combination in which work done is maximum

(A) $(I)\ (i)\ (R)$

(B) $(II)\ (i)\ (P)$

(C) $(III)\ (iv)\ (S)$

(D) $(III)\ (ii)\ (R)$

Space for Rough work

12. Pick a combination in which thermal energy supplied is maximum
(A) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$
(B) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{S})$
(C) $(\mathrm{I})\ (\mathrm{i})\ (\mathrm{R})$
(D) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{Q})$

13. Pick a correct combination
(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{P})$
(B) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{Q})$
(C) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{S})$
(D) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{Q})$

SECTION – B
(Single digit integer type)

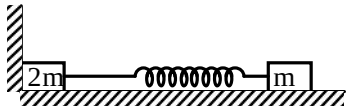
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. A container filled with air under pressure P_0 contains a soap bubble of radius *R*. The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S, P_0 is found to be $\frac{16nS}{R}\mathrm{SI\ unit}$. Find *n*.

15. In an L–C circuit, oscillations of energy *E* are sustained. The capacitor plates are now slowly drawn apart to increase the oscillation frequency three times. The work done in the process is *nE*. Find *n*.

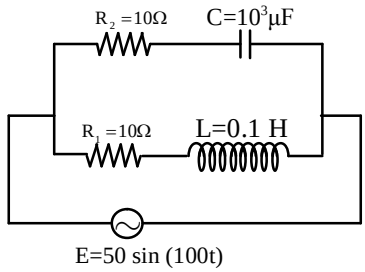
Space for Rough work

16. Two blocks of mass *m* and 2m connected by a weightless spring of stiffness *k* rest on a smooth horizontal plane. Block of mass *m* is shifted to a small distance *x* to the left and then released. If the duration between releasing mass *m* and breaking off of mass 2m is Δt, then average force(in newton) exerted by wall on mass 2m is given as $\frac{x}{\Delta t}\sqrt{mkn}$. Find the value of *n*.



17. One end of a string of length *L* = 15m is tied to the ceiling of an elevator accelerating upwards with an acceleration $20\ \text{m/s}^2$. The other end of the string is free. The linear mass density of the string varies linearly from 0 to λ_0 from bottom to top. Find the time taken by a pulse (in seconds) to reach from bottom to the top. $\left(\text{g= } 10\ \text{ms}^{-2}\right)$

18. For the circuit shown in the figure, find the peak current (in ampere) through the source.



Space for Rough work

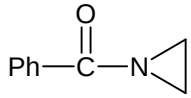
Chemistry

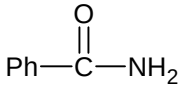
PART – II

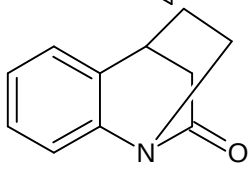
SECTION – A
(One or More than one correct type)

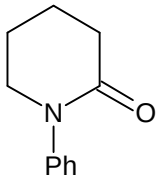
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. In the following compound

(i) 

(ii) 

(iii) 

(iv) 

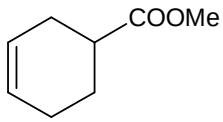
The rate of alkaline hydrolysis of the amide:

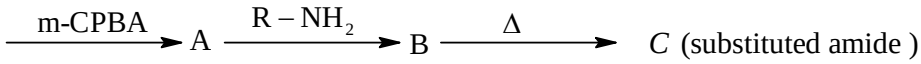
(A) I>II

(B) III>IV

(C) III<IV

(D) I<II

20. 



R=Methyl

(A) The compound 'A' has a three membered ring.

(B) Degree of unsaturation in C is 3

(C) In 'B', $-NHR$ and OH group are Trans to each other

(D) In B, $-COOMe$ and OH group are Trans to each other.

Space for Rough work

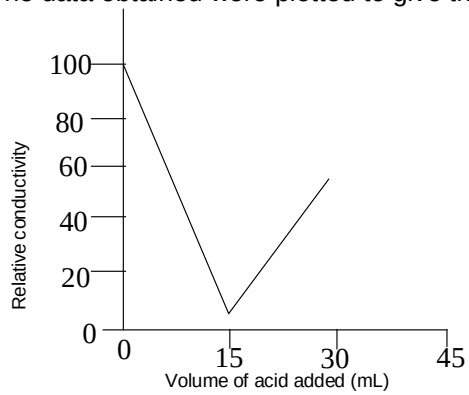
21. Identify the incorrect statement (s)?
(A) The ratio of the observed molar volume to ideal volume under similar condition is called compressibility factor.
(B) The value of van der Waals constants 'a' and 'b' is independent of the nature of gas.
(C) At high temperature and low pressure, $\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$ is used to explain the behaviour of any gas.
(D) H_2 and He gases are difficult to liquefy because their Critical temperature is very small.

22. In which of the following reactions, the stiochiometric ratio of H_2SO_4 to H_2 gas is the same?
(A $Zn(s)$ and H_2SO_4
(B) $Na(s)$ and H_2SO_4
(C) $Al(s)$ and *conc*.H_2SO_4
(D) $Cu(s)$ and H_2SO_4

23. A 10 mL mixture containing 0.2 g NaOH and $0.25\,g\ Na_2CO_3$ was treated with $x\,N\ HCl$ required 15 mL of it with phenolphthalein indicator. After end point a $0.5\ g\ NaHCO_3$ was added and it was again titrated with $y\ N\ HCl$ required 25.5 mL of HCl with methyl orange. Which of the following is/are correct?
(A) x is almost 1.5 times of y
(B) If in above titration x N HCl is replaced with y N HCl then for first end point it will require 23mL of HCl.
(C) After first end point instead of 0.5 g if 2.5 g $NaHCO_3$ is added and fitrated with $x\ N\ HCl$ it will require 65.5 mL.
(D) Only (A) and (B) are correct.

Space for Rough work

24. $20\,\text{mL}$ of KOH solution was titrated with $0.20\,\text{mol L}^{-1}$ H_2SO_4 solution in a conductively cell. The data obtained were plotted to give the graph shown below:



The concentration of the KOH solution was:

(A) $0.30\,\text{mol L}^{-1}$

(B) $0.15\,\text{mol L}^{-1}$

(C) $0.12\,\text{mol L}^{-1}$

(D) $0.075\,\text{mol L}^{-1}$

25. For the reaction $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$. If Kp of the above reaction is 78.1 at $400°\text{C}$ which is not correct:

(A) At pressure P degree of dissociation of NH_3 is $\left[1+\frac{3\sqrt{3}P}{4k_p}\right]^{-1/2}$

(B) $K_c = 1.413\,mol/\ell$

(C) If we increase pressure it will decrease K

(D) If we increase P by bringing piston down, then, degree of dissociation α will decrease

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

	Column -1 (Ores)		Column -2 (Type of Ore)		Column -3 (Process involved in extraction/refining)
(I)	Dolomite	(i)	Chloride	(P)	Self reduction
(II)	Cassiterite	(ii)	Carbonate	(Q)	Carbon reduction
(III)	Chalcocite	(iii)	Sulphide	(R)	Fused electrolysis
(IV)	Azurite	(iv)	Oxide	(S)	Poling

26. Which of the following is suitable for calcium?

(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{P})$

(B) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{R})$

(C) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{S})$

(D) $(\mathrm{III})\ (\mathrm{ii})\ (\mathrm{R})$

27. Which is correct for tin?

(A) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

(B) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{R})$

(C) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{Q})$

(D) $(\mathrm{IV})\ (\mathrm{ii})\ (\mathrm{Q})$

Space for Rough work

28. Which of the following is used for extraction of copper?

(A) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{R})$

(B) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{Q})$

(C) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{P})$

(D) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{Q})$

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Consider all the gases as ideal and irreversible process is carried out at constant P_{Final}.

	Column -1		Column -2		Column -3
(I)	A(g) (10atm, 1mol, 300K,V, ideal gas) $\rightarrow$ A(g) (1atm, 1mol, 10V)	(i)	q=0	(P)	$\Delta U=0$
(II)	$B(s) \xrightarrow{\text{At } 0^0\text{C, 1atm.}} B(liq)$ (melting)	(ii)	W=0	(Q)	$\Delta H=0$
(III)	A(g) (10atm, 1mol, 300K, $C_v=1.5R$) $\rightarrow$ A(g) (1atm, 1mol)	(iii)	$\Delta S_{system}>0$	(R)	$Vol._{Final} < Vol._{Inital}$
(IV)	Mixing of ideal gases at constant T and P in an isolated container	(iv)	$\Delta G=0$	(S)	$T_{Final\ for\ irr.\ process} > T_{Final\ for\ rev.\ process}$

29. Which of the following combination represents isothermal reversible process?

(A) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{Q})$

(B) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{S})$

(C) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$

(D) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{P})$

Space for Rough work

30. Which of the following is correct combination when "B" as H_2O and others are gases as specified?

(A) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{Q})$

(B) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{R})$

(C) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{S})$

(D) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{S})$

31. Which of the following combination represents the adiabatic process?

(A) $(\mathrm{III})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{III})\ (\mathrm{i})\ (\mathrm{Q})$

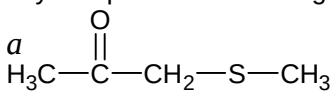
(C) $(\mathrm{III})\ (\mathrm{i})\ (\mathrm{S})$

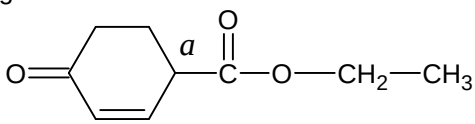
(D) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{S})$

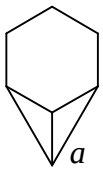
SECTION – B
(Single digit integer type)

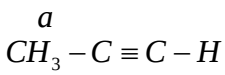
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

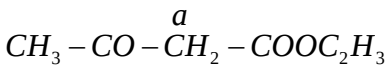
32. How many compounds are having most acidic hydrogen attached to the carbon indicated as (a)?

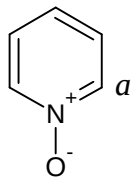
(i) 

(ii) 

(iii) 

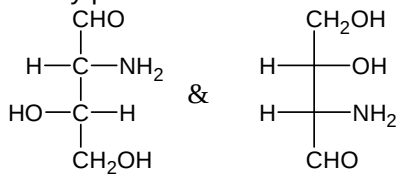
(iv) 

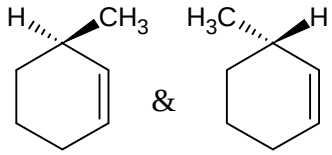
(v) 

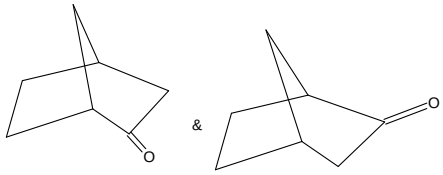
(vi) 

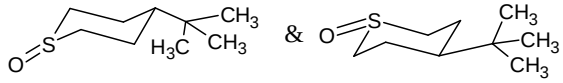
Space for Rough work

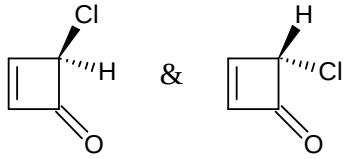
33. How many pairs are Enantiomers?

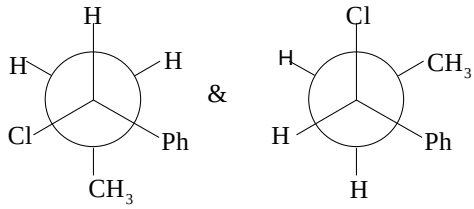
(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

34. How many statements are true?

(a) At constant T and P, $\Delta_{mix}G$ must be negative for an ideal solution.

(b) At constant T and P, $\Delta_{mix}G$ must be negative for every solution.

(c) At constant T and P, $\Delta_{mix}S=\Delta_{mix}H/T$ for an ideal solution.

(d) For equilibrium between a solution and its vapor, G_m of the solution must equal G_m of the vapor.

(e) For equilibrium between an ideal solution and an ideal vapor x_B^l must equal x_B^v.

(f) In an ideal solution, the partial molar volume of a component equals the molar volume of the pure substance.

35. Positronium is a species consisting of an electron bound to a positron. The ionization potential of positronium is 'A' ev, the value of $\frac{A}{3.4}$ will be:

36. Choose the correct reactions?

(i) When $CuSO_4$ reacts with NH_3 red color complex is formed

(ii) When $CuSO_4$ reacts with PH_3 , Cu_3P is formed

(iii) $NH_3(\text{excess})+Cl_2 \longrightarrow NH_4Cl+N_2$

(iv) $NH_3+Cl_2(\text{excess}) \longrightarrow NCl_3+HCl$

(v) $HNO_3+P_4O_{10} \xrightarrow{\Delta} HPO_3+N_2O_5$

(vi) $SbF_5+XeF_4 \longrightarrow [SbF_4]^+[XeF_5]^-$

(vii) $XeF_6+SiO_2 \longrightarrow XeOF_4+SiF_4$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If the equation $x^3 + 2x^2 + x + k = 0$ has three real roots in which two roots are equal, then value of k is/are

(A) 0 (B) $k = \frac{27}{4}$

(C) $k = \frac{4}{27}$ (D) None of these

38. For a given parabola $y^2 = 4ax$, two variable chords PQ and RS at right angles are drawn through the fixed point $A(x_1, y_1)$ inside the parabola, making variable angles θ and α with x–axis. If r_1, r_2, r_3, r_4 are distances of P, Q, R and S from A, then the value of $\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4}$

(A) Independent of θ (B) Independent of α
(C) Depends upon both θ and α (D) Is a constant

39. From (1, 2), tangents are drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$. Then
(A) Sum of x–coordinates of points of contact is zero
(B) Sum of x–coordinates of points of contact is 4
(C) Sum of y–coordinates of points of contact is zero
(D) Sum of y–coordinates of points of contact is 4

40. If $\left|\int_a^b f(x)dx\right| = \int_a^b |f(x)|dx \ \ (a \neq b)$ also $f(x) \neq 0$ for any $x \in (a, b)$ and $g(x) = \int_0^x f(x)dx$, then $\int_a^b f(x)g(x)dx$ when a and b are positive

(A) Cannot be positive (B) Cannot be negative
(C) Cannot be equal to zero (D) Nothing can be said

Space for Rough work

41. If $\alpha, \beta, \gamma \in R$, then the determinant

$$\Delta = \begin{vmatrix} (e^{i\alpha} + e^{-i\alpha})^2 & (e^{i\alpha} - e^{-i\alpha})^2 & 4 \\ (e^{i\beta} + e^{-i\beta})^2 & (e^{i\beta} - e^{-i\beta})^2 & 4 \\ (e^{i\gamma} + e^{-i\gamma})^2 & (e^{i\gamma} - e^{-i\gamma})^2 & 4 \end{vmatrix} \text{ is}$$

(A) Equal to 0
(B) Equal to $e^{i(\alpha+\beta+\gamma)}$
(C) Independent of α, β and γ
(D) Dependent of α, β and

42. If AB = A and BA = B, where A and B are square matrices, then
(A) $A^2 = A$
(B) $A^2 = B$
(C) $B^2 = B$
(D) $B^2 = A$

43. If a complex number satisfies the equation $|z| - \log_2(z + 2) = 0$ (Re z > 0), then
(A) $z = -\bar{z}$
(B) $|z| = 2$
(C) $\arg(z) = \frac{\pi}{2}$
(D) $\arg(z) = 0$

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

	Column -1		Column -2		Column -3
(I)	The value of $\sum_{n=0}^{\infty} 2^{-n+(-1)^n}$ is	(i)	The value of $\left[\sum_{n=0}^{\infty}\tan^{-1}\left(\frac{2}{\sqrt{n+2}+\sqrt{n}+(n+2)\sqrt{n}+n\left(\sqrt{n+2}\right)}\right)\right]$ (where [.] denotes greatest integer function) is	(P)	2
(II)	Number of solutions of equation $16\sin^3 x = 14 + \sqrt[3]{\sin x + 7}$ in $[0, 4\pi]$ is	(ii)	If a, b, c are sides of a triangle, then $\frac{2\left(a^2+b^2+c^2\right)}{ab+bc+ca}$ is always less than	(Q)	3
(III)	Number of solutions of equation $\tan^{-1}(2\sin x) = \cot^{-1}(\cos x)$ in $[0, 10\pi]$ is	(iii)	If a, b, c are distinct real numbers such that $a^2(b+c) = b^2(a+c) = 2$, then the value of $c^2(a+b)$ is	(R)	4
(IV)	If the sum of roots of the quadratic equation $(-a+1)x^2 + (a^2 - a + 4)x - 2a + 3 = 0$ is minimum, then the value of a(a>1) is	(iv)	If $P=\sin A\sin B, Q=\sin C\cos A,$ $R=\sin A\cos B$ $S=\cos A\cos C$, then the value of $5\left(P^2+Q^2+R^2+S^2\right)$ is	(S)	5

44. Which of the following options is the only **CORRECT** combination?

(A) $(I)\ (ii)\ (R)$

(B) $(II)\ (iv)\ (S)$

(C) $(III)\ (i)\ (Q)$

(D) $(IV)\ (iv)\ (S)$

Space for Rough work

45. Which of the following options is the only **CORRECT** combination?

(A) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{R})$

(C) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{P})$

(D) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{Q})$

46. Which of the following options is the only **CORRECT** combination?

(A) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{R})$

(C) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

(D) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$

Space for Rough work

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

	Column -1		Column -2		Col. -3
(I)	If $S=\{Z:\|Z+\bar{Z}\|+2\|Z-\bar{Z}\|=4$ and $\|Z\|$ is minimum$\}$ then $\frac{25}{8}A$ is (where A is area of polygon formed of all points in S taking as vertices)	(i)	If the Point $(\sec\alpha, \operatorname{cosec}\alpha)$ moves in the plane of circle $x^2+y^2=3$ and the minimum distance of this point from circle is $a-\sqrt{b}(a,b\in N)$ then $a+b$	(P)	2
(II)	Let $S=\{Z\|Z\bar{Z}-(3-4i)Z-(3+4i)\bar{Z}+21=0\}$ If M and m be maximum value and minimum value of $\frac{Z-\bar{Z}}{i(Z+\bar{Z})}$ then $\frac{1}{M}+\frac{1}{m}$ is	(ii)	Two circles $x^2+y^2+2n_1x+2y+\frac{1}{2}=0$ and $x^2+y^2+n_2x+n_2y+n_1=\frac{1}{2}$ intersect each other orthogonally where n_1, n_2 are integers then the number of possible ordered pairs (n_1,n_2) is	(Q)	3
(III)	Let x is the minimum value of $\|Z\|^2+\|Z-3\|^2+\|Z-6i\|^2$ then $\frac{x}{10}$ is	(iii)	If $a_n=\sqrt{1+\left(1+\frac{1}{n}\right)^2}+\sqrt{1+\left(1-\frac{1}{n}\right)^2}$, then the value of $\left(\sum_{n=1}^{20}\frac{1}{a_n}\right)-3$ is	(R)	4
(IV)	Consider a triangle formed by the points $A\left(\frac{2}{\sqrt{3}}e^{i(\pi/2)}\right), B\left(\frac{2}{\sqrt{3}}e^{-i\left(\frac{\pi}{6}\right)}\right), C\left(\frac{2}{\sqrt{3}}e^{-i\left(\frac{5\pi}{6}\right)}\right)$ Let P(Z) is any point on it's in-circle, then $AP^2+BP^2+CP^2$ is	(iv)	The ea^n $9x^3+9x^2y-45x^2=4y^3+4xy^2-20y^2$ represents 3 straight lines two of which pass through origin then $\frac{1}{10}$ (Area of triangle formed by these lines)	(S)	5

47. Which of the following options is the only **CORRECT** combination?

(A) $(I)\ (ii)\ (R)$

(B) $(II)\ (iv)\ (S)$

(C) $(III)\ (i)\ (Q)$

(D) $(IV)\ (iv)\ (S)$

Space for Rough work

48. Which of the following options is the only **CORRECT** combination?

(A) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{R})$

(C) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{P})$

(D) $(\mathrm{IV})\ (\mathrm{iii})\ (\mathrm{Q})$

49. Which of the following options is the only **INCORRECT** combination?

(A) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{R})$

(C) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{P})$

(D) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. Find number of positive integral solution of equation $\tan^{-1} x + \cos^{-1} \dfrac{y}{\sqrt{1-y^2}} = \sin^{-1} \dfrac{3}{\sqrt{10}}$

51. Tangent are drawn from any point on the hyperbola $x^2/9 - y^2/4 = 1$ to the circle $x^2 + y^2 = 9$. If the locus of the mid-point of the chord of contact is a $a\left(x^2+y^2\right)^2 = bx^2 - cy^2$, then the value of $a^2 + b^2 + c^2 - 7870$ is equal to……

Space for Rough work

52. If $A=\begin{pmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{pmatrix}$ and $A^2 - 4A - 5I = O$ where I and O are the unit matrix and the null matrix of order 3 respectively. If $15A^{-1}=\lambda\begin{vmatrix}-3 & 2 & 2\\ 2 & -3 & 2\\ 2 & 2 & -3\end{vmatrix}$ then the find the value of λ.

53. The plane $2x-2y+z=3$ is rotated about the line where it cuts the xy plane by an acute angle α. If the new position of plane contains the point (3, 1, 1) then $9\cos\alpha$ equal to:

54. If $0 < \alpha, \beta < 2\pi$, then the number of ordered pairs (α, β) satisfying $\sin^2(\alpha+\beta) - 2\sin\alpha\sin(\alpha+\beta) + \sin^2\alpha + \cos^2\beta = 0$ is

Space for Rough work